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Stroke

ORIGINAL CONTRIBUTION

Poorer Outcomes With Blood Pressure Lowering in Severe White Matter Disease After Thrombectomy

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BACKGROUND: Intensive blood pressure (BP) management after successful endovascular thrombectomy (EVT) worsens outcomes in acute ischemic stroke. White matter hyperintensity (WMH), associated with impaired cerebral autoregulation, is associated with poor outcomes after EVT; however, its role in modifying BP management effects remains unclear. We investigated whether WMH severity modifies the effect of post-EVT BP management on outcomes.

METHODS: This secondary analysis of the OPTIMAL-BP trial (Outcome in Patients Treated With Intra-arterial Thrombectomy–Optimal Blood Pressure Control; conducted at 19 South Korean stroke centers, 2020–2022) included patients with available Fluid-Attenuated Inversion Recovery images. The trial compared intensive (<140 mm Hg) versus conventional (140–180 mm Hg) systolic BP management for 24 hours post-EVT. WMH severity was assessed via the Fazekas scale (severe defined as a total score ≥3). The primary outcome was functional independence (modified Rankin Scale score, 0–2) at 3 months. Safety outcomes included symptomatic intracerebral hemorrhage within 36 hours and stroke-related death within 3 months. Multivariable logistic regression was used to estimate adjusted odds ratios and 95% CI, adjusting for age, sex, onset-to-reperfusion time, intravenous thrombolysis, complete reperfusion, and baseline stroke severity.

RESULTS: Of 271 patients (median, 75.0 years; 59.0% men), 163 (60.1%) had severe WMH. In this group, intensive management was associated with a lower rate of functional independence compared with conventional management (29.9% versus 51.3%; adjusted odds ratio, 0.36 [95% CI, 0.16–0.79]; P=0.011). In contrast, no significant difference was observed in patients with mild WMH (55.8% versus 69.6%; adjusted odds ratio, 0.77 [95% CI, 0.31–1.95]; P=0.584). Symptomatic intracerebral hemorrhage and stroke-related death rates did not differ by BP management in either group.

CONCLUSIONS: Intensive BP management post-EVT was associated with poorer functional outcomes, particularly in patients with severe WMH. WMH burden may modify BP control effects, supporting individualized BP targets based on WMH severity.

REGISTRATION: URL: https://clinicaltrials.gov/study; Unique identifier: NCT04205305.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: blood pressure ■ ischemic stroke ■ magnetic resonance imaging ■ thrombectomy ■ white matter

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Nonstandard Abbreviations and Acronyms

aOR	adjusted odds ratio
BP	blood pressure
CONSORT	Consolidated Standards of Reporting Trials
EVT	endovascular thrombectomy
FLAIR	fluid-attenuated inversion recovery
MRI	magnetic resonance imaging
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
OPTIMAL-BP	Outcome in Patients Treated With Intra-arterial Thrombectomy–Optimal Blood Pressure Control
WMH	white matter hyperintensity

Stroke guidelines recommend maintaining blood pressure (BP) below 180/105 mm Hg for 24 hours after successful endovascular thrombectomy (EVT) in patients with acute ischemic stroke due to large vessel occlusion.[1,2] However, this recommendation is based primarily on expert opinion.[1,2] Observational studies have suggested that intensive BP management may reduce the risk of hemorrhagic transformation and associated poor outcomes.[3,4] Recent randomized trials and a meta-analysis have challenged this notion, showing that intensive BP management was not beneficial and may increase harm without reducing the risk of hemorrhagic transformation.[5–9]

White matter hyperintensity (WMH) in the brain is a manifestation of cerebral small vessel disease, appearing as signal abnormalities of variable size in the white matter, typically hyperintense on fluid-attenuated inversion recovery (FLAIR) image of magnetic resonance imaging (MRI).[10,11] In the presence of WMH, impaired autoregulation may render the ischemic brain more vulnerable to BP fluctuations,[12] with intensive BP management in the early reperfusion phase potentially reducing cerebral perfusion pressure and exacerbating hypoperfusion-related injury.[7] However, no previous study has investigated whether the effect of intensive BP management after EVT differs according to the WMH burden.

We hypothesized that patients with severe WMH would experience worse functional outcomes under intensive BP lowering due to impaired cerebral autoregulation. Therefore, we conducted a secondary analysis of the OPTIMAL-BP trial (Outcome in Patients Treated With Intra-Arterial Thrombectomy–Optimal Blood Pressure Control) to investigate the differential effect of intensive BP management according to the severity of WMH in patients with successful reperfusion after EVT.

METHODS

This study was a secondary analysis of the OPTIMAL-BP trial, not prespecified in the original protocol. The OPTIMAL-BP trial was approved by the ethics committee at Severance Hospital (approval number: 4-2019-1208) and all participating centers. This study is reported according to CONSORT guidelines (Consolidated Standards of Reporting Trials). Written informed consent was obtained from participants or their legally authorized representatives before the study began. The data that support the findings of this study are available from the corresponding author on reasonable request. Dr Nam had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study Design and Participants

The OPTIMAL-BP trial was a multicenter, randomized, open-label, blinded-end point trial conducted at 19 stroke centers in South Korea from June 2020 to November 2022.[7] The details of the protocol, including detailed inclusion and exclusion criteria, statistical analysis plan, and the results of the trial, were reported elsewhere.[7,13] Briefly, the trial enrolled patients with acute ischemic stroke due to large vessel occlusion who achieved successful reperfusion of the occluded artery and exhibited elevated systolic BP (≥140 mm Hg) within 2 hours after reperfusion. Patients with prestroke disability (modified Rankin Scale [mRS] score, 3–5) were excluded. After successful EVT, patients were randomly allocated to intensive management (target systolic BP <140 mm Hg) or to conventional management (target systolic BP, 140–180 mm Hg). During the trial, patients were managed in accordance with the prevailing stroke guidelines at that time.[1,14] For this secondary analysis, we screened intention-to-treat patients included in the primary analysis of the OPTIMAL-BP trial and further excluded patients without an MRI or FLAIR image.

WMH Rating

WMH was defined as a supratentorial hyperintense lesion on FLAIR imaging, in accordance with the Standards for Reporting Vascular Changes on Neuroimaging criteria.[10] The brain MRI examinations were performed using either a 1.5T or 3.0T MRI system, following local protocols during admission for EVT. For WMH assessment, a hierarchical imaging selection protocol was implemented to ensure standardized evaluation across the cohort. FLAIR images acquired before EVT were prioritized to represent the baseline premorbid state. In cases where preprocedural MRI was unavailable due to the emergency nature of the stroke, the earliest obtainable post-EVT FLAIR scans were used. To mitigate potential confounding from lesion evolution, the first chronological FLAIR image was selected for rating in all patients. To ensure diagnostic accuracy, all assessments were cross-referenced with diffusion-weighted imaging to strictly differentiate chronic WMH from acute ischemic lesions. Final infarct volume was quantified exclusively using diffusion-weighted imaging performed after the EVT procedure.

The degrees of both periventricular WMH and deep WMH were assessed using a 4-point ordinal scale based on the Fazekas method.[15] Total WMH score was calculated as the sum of periventricular and deep WMH scores, ranging from 0 to 6. The main independent variable in this study was the

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dichotomized total WMH score: mild (0–2) and severe (3–6).[16] For supplemental analyses, the total WMH scores were further categorized into 3 groups: 0 to 2, 3 to 4, and 5 to 6. Each of the periventricular WMH and deep WMH scores was dichotomized into 0 to 1 and 2 to 3.

All imaging data were submitted to an independent core laboratory for blinded adjudication. Two raters, blinded to clinical information, independently assessed WMH severity. Interrater reliability was excellent, with a quadratic weighted kappa value of 0.815 (Figure S1).

Outcome Measures

The primary efficacy outcome was functional independence at 3 months, defined by a binary classification of the mRS score 0 to 2, indicating independence, and 3 to 6, indicating dependence or death. Functional outcomes at 3 months were assessed by trained medical personnel who were blinded to treatment allocation and imaging data, and evaluations were conducted either in person or via telephone. The mRS score was determined through a clinician-rated assessment. This assessment was conducted by certified medical staff, blinded to treatment allocation, who followed a structured interview guide based on the validated Korean version of the mRS (http://stroke-edu.or.kr). The primary safety outcomes included symptomatic intracerebral hemorrhage within 36 hours and mortality related to the index stroke within 3 months. Symptomatic intracerebral hemorrhage was defined based on the European Cooperative Acute Stroke Study III criteria as the presence of extravascular blood in the brain or cranium accompanied by neurological deterioration, indicated by an increase of ≥4 points in the National Institutes of Health Stroke Scale (NIHSS) score or resulting in death.[17] Secondary outcomes included a shift analysis of the full distribution of mRS scores at 3 months; excellent recovery on the NIHSS score at 24 hours, defined as an NIHSS score of 0 to 1 or an improvement of >8 points; functional independence at 1 month (mRS score, 0–2); health-related quality of life assessed by the EuroQoL 5-Dimension, 3-Level Questionnaire; and infarct volume (milliliter).

Statistical Analyses

Group comparisons were performed using the χ^2 test for categorical variables and the *t* test or Mann-Whitney *U* test for continuous variables, as appropriate. To assess the effect of intensive versus conventional BP management on the primary outcome, binomial logistic regression was conducted, and results were reported as adjusted odds ratios (aORs) with 95% CIs. The model was adjusted for age, sex, time from onset to reperfusion, IV thrombolysis, modified Thrombolysis in Cerebral Infarction 3 recanalization, and NIHSS score immediately before EVT, in line with the OPTIMAL-BP trial design with minor modifications.[7] Secondary outcomes were analyzed using ordinal logistic, binary logistic, or linear regression models, adjusted for the same covariates as the primary analysis. Interaction terms were included to evaluate potential effect modification. For supplemental analyses, the effect of intensive versus conventional BP management on the primary efficacy outcome was further evaluated according to the trichotomized total WMH scores (0–2, 3–4, and 5–6) and the dichotomized periventricular WMH and deep WMH scores (0–1 and 2–3). All statistical analyses were performed using R (version 4.3.1; R Foundation for Statistical Computing), and statistical significance was defined as a 2-sided $P<0.05$.

RESULTS

Among the 302 patients included in the OPTIMAL-BP trial, 11 patients were excluded due to the absence of MRI, 18 due to the lack of FLAIR imaging, and 2 due to inadequate FLAIR image quality (Figure 1). Consequently, a total of 271 patients were included (median, 75.0 years; 59.0% men). Severe WMH was observed in 163 patients (60.1%)—87 in the intensive management group and 76 in the conventional management group—while mild WMH was observed in 108 patients (39.9%)—52 in the intensive group and 56 in the conventional group. Baseline characteristics—including demographics, comorbidities, radiological and procedural variables, time parameters, and BP at randomization did not significantly differ between the intensive and conventional management groups in patients with either severe or mild WMH (Table 1). In both WMH severity groups, mean systolic BP during the first 24 hours was significantly lower in the intensive management group compared with the conventional group ($P<0.05$; Table 1; Figure S2).

Primary Efficacy Outcome

Among patients with severe WMH, the proportion achieving functional independence at 3 months was significantly lower in the intensive management group (26/87, 29.9%) compared with the conventional management group (39/76, 51.3%). Multivariable logistic regression analysis showed that intensive management was associated with a 64% lower odds of achieving functional independence (aOR,0.36 [95% CI, 0.16–0.79]; $P=0.011$; Figure 2; Table 2). In contrast, among patients with mild WMH, no significant difference was observed between the 2 management groups (intensive: 29/52, 55.8%; conventional group: 39/56, 69.6%; aOR, 0.77 [95% CI, 0.31–1.95]; $P=0.584$; Figure 2; Table 2).

Primary Safety Outcomes

No significant differences in the primary safety outcomes—symptomatic intracerebral hemorrhage and stroke-related mortality within 3 months—were observed between the intensive and conventional management groups in patients with either severe or mild WMH (all $P>0.05$; Table 2).

Secondary Outcomes

In patients with severe WMH, the mRS shift analysis demonstrated worse scores in the intensive management group (aOR, 0.50 [95% CI, 0.28–0.88]; $P=0.015$; Table 2). Similarly, the EuroQoL 5-Dimension, 3-Level

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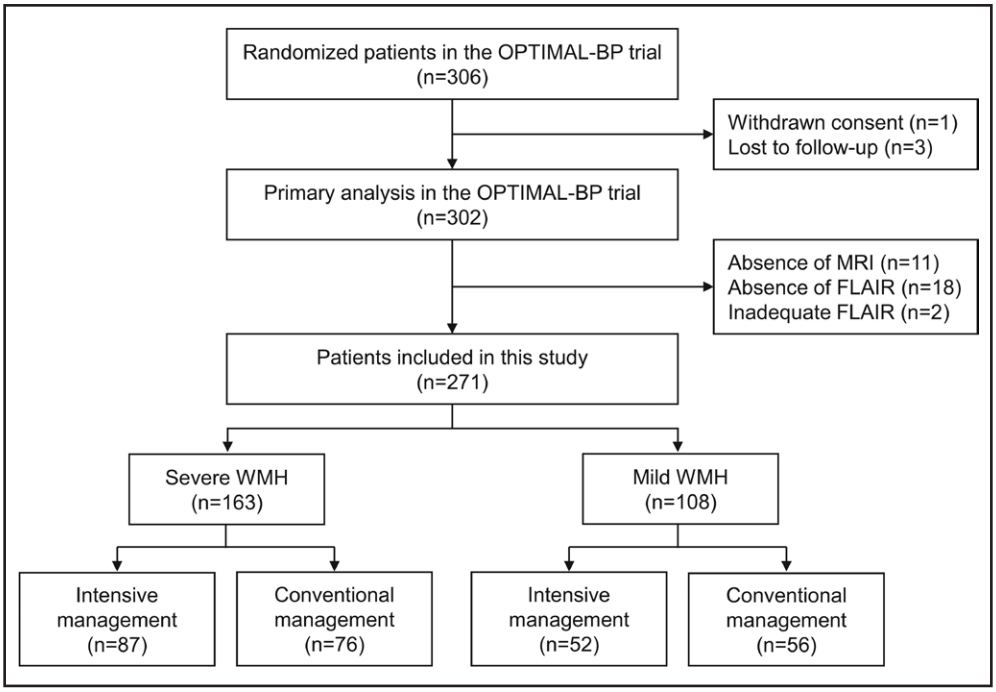

Figure 1. Flow diagram of patient inclusion.
FLAIR indicates fluid-attenuated inversion recovery; OPTIMAL-BP, Outcome in Patients Treated With Intra-Arterial Thrombectomy–Optimal Blood Pressure Control; MRI, magnetic resonance imaging; and WMH, white matter hyperintensity.

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Questionnaire score was significantly lower in the intensive management group (0.4) compared with the conventional group (0.7), with an adjusted mean difference of −0.12 (95% CI, −0.23 to −0.01; *P*=0.037; Table 2). However, among patients with mild WMH, no significant differences in these secondary outcomes were observed between the 2 management groups (all *P*>0.05; Table 2).

Furthermore, no significant differences were found between the intensive and conventional groups in other secondary outcomes—including excellent recovery in NIHSS score at 24 hours, and functional independence at 1 month—among patients with either severe or mild WMH (all *P*>0.05; Table 2).

In patients with mild WMH, the infarct volume was significantly larger in the intensive management group (19.8 mL) than in the conventional group (12.9 mL; adjusted mean difference, 29.69 mL [95% CI, 0.07–59.32]; *P*=0.049; Table 2); whereas no significant difference was observed among patients with severe WMH (*P*>0.05; Table 2).

Sensitivity Analyses

In a supplemental analysis using trichotomized total WMH score groups, patients with scores of 5 to 6 had a significantly lower odds of functional independence at 3 months under intensive management (aOR, 0.04 [95% CI, 0.00–0.69]; *P*=0.027). A borderline association was noted in the 3 to 4 group (aOR, 0.43 [95% CI, 0.18–1.04]; *P*=0.060), while no significant difference was found in the 0 to 2 group (aOR, 0.77 [95% CI, 0.31–1.95]; *P*=0.584; Figure S3; Table S1). In further supplemental analyses evaluating dichotomized periventricular and deep WMH separately, intensive management was associated with a significantly lower odds of functional independence in patients with periventricular WMH scores of 2 to 3 (aOR, 0.33 [95% CI, 0.15–0.75]; *P*=0.008) and deep WMH scores of 0 to 1 (aOR, 0.47 [95% CI, 0.23–0.96]; *P*=0.037); however, no significant association was observed in patients with periventricular WMH scores of 0 to 1 or deep WMH scores of 2 to 3 (Figures S4 and S5; Table S2).

DISCUSSION

In this secondary analysis of the OPTIMAL-BP trial, intensive BP management during the first 24 hours after successful EVT was associated with significantly lower odds of achieving functional independence at 3 months in patients with severe WMH, but not in those with mild WMH. These findings suggest that the burden of WMH may modify the effect of post-EVT BP management on functional recovery.

To explain the observed harm of intensive BP lowering in patients with high WMH burden after EVT, we considered underlying pathophysiological mechanisms. In these patients, impaired cerebral autoregulation, known to be associated with high WMH burden, may result in reduced cerebral blood flow and diminished vascular reactivity, thereby increasing susceptibility to hypoperfusion.[12,18]

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Table 1. Baseline Characteristics

	Severe WMH			Mild WMH		
	Intensive management (n=87)	Conventional management (n=76)	*P* value	Intensive management (n=52)	Conventional management (n=56)	*P* value
Demographics and medical condition						
Age, median (IQR), y	80.0 (73.5–83.5)	79.0 (70.5–83.0)	0.517	67.0 (57.0–78.0)	67.5 (59.0–75.5)	0.834
Sex (men), n (%)	45 (51.7)	39 (51.3)	>0.999	37 (71.2)	39 (69.6)	>0.999
Hypertension	68 (78.2)	56 (73.7)	0.628	39 (75.0)	42 (75.0)	>0.999
Atrial fibrillation	41 (47.1)	42 (55.3)	0.379	25 (48.1)	20 (35.7)	0.268
Diabetes	33 (37.9)	32 (42.1)	0.702	23 (44.2)	19 (33.9)	0.368
Hyperlipidemia	30 (34.5)	31 (40.8)	0.504	24 (46.2)	17 (30.4)	0.136
Smoking	20 (23.0)	12 (15.8)	0.339	17 (32.7)	14 (25.0)	0.503
Previous stroke	22 (25.3)	17 (22.4)	0.801	11 (21.2)	9 (16.1)	0.666
Coronary artery obstructive disease	11 (12.6)	12 (15.8)	0.726	5 (9.6)	4 (7.1)	0.908
Active cancer	6 (6.9)	2 (2.6)	0.371	2 (3.8)	3 (5.4)	>0.999
Congestive heart failure	3 (3.4)	4 (5.3)	0.855	2 (3.8)	3 (5.4)	>0.999
Prestroke mRS score <2	72 (82.8)	66 (86.8)	0.614	51 (98.1)	55 (98.2)	1.000
NIHSS score immediately before EVT	13.0 (9.0–18.5)	11.0 (7.0–17.5)	0.225	12.5 (8.0–16.5)	9.0 (6.0–15.0)	0.080
IV-tPA	25 (28.7)	26 (34.2)	0.560	13 (25.0)	23 (41.1)	0.117
Radiological and procedural variables						
Site of occlusion in the anterior circulation	78 (89.7)	69 (90.8)	>0.999	47 (90.4)	52 (92.9)	0.908
ASPECTS ≥6	82 (96.5)	73 (96.1)	>0.999	47 (92.2)	55 (98.2)	0.306
Good collateral	54 (69.2)	39 (53.4)	0.068	38 (76.0)	42 (79.2)	0.874
mTICI immediate			0.322			0.081
2b	22 (25.3)	12 (15.8)		8 (15.4)	15 (26.8)	
2c	9 (10.3)	8 (10.5)		7 (13.5)	13 (23.2)	
3	56 (64.4)	56 (73.7)		37 (71.2)	28 (50.0)	
Time parameters, median (IQR)						
Onset to puncture, min	440.0 (221.0–833.5)	358.5 (197.0–786.5)	0.518	358.5 (222.5–666.5)	312.0 (203.0–725.0)	0.726
Puncture to reperfusion, min	30.0 (23.0–44.5)	30.0 (19.0–46.5)	0.425	30.5 (18.5–47.0)	31.5 (21.5–44.5)	0.775
Onset to randomization, min	540.0 (327.5–955.0)	481.0 (285.5–885.0)	0.382	480.0 (320.0–752.5)	462.5 (312.5–825.0)	0.966
Onset to reperfusion, min	468.0 (260.5–873.5)	391.0 (235.5–801.0)	0.523	407.0 (250.5–691.0)	369.5 (254.5–757.5)	0.731
Onset to FLAIR, day	1.2 (0.5–1.9)	1.1 (0.4–1.6)	0.301	0.9 (0.3–1.5)	1.1 (0.5–1.9)	0.408
BP variables, mean±SD, mm Hg						
SBP at randomization	155.8±13.9	154.8±11.0	0.609	155.4±13.6	155.9±19.1	0.877
Mean SBP for 24 h	129.3±7.8	140.9±13.1	<0.001	129.7±6.9	135.1±13.8	0.010

ASPECTS indicates Alberta Stroke Program Early CT Score; BP, blood pressure; EVT, endovascular thrombectomy; FLAIR, Fluid Attenuated Inversion Recovery; IQR, interquartile range; IV-tPA, intravenous tissue-type plasminogen activator; mRS, modified Rankin Scale; mTICI, modified Treatment in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; SBP, systolic BP; and WMH, white matter hyperintensity.

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Although our study did not directly assess the degree of impaired cerebral autoregulation, previous studies have demonstrated a correlation between WMH burden and impaired cerebral perfusion.[19,20] Despite successful macrovascular recanalization, ischemic regions—particularly within the penumbra—may remain in an oligemic state.[21–23] In individuals with extensive WMH, small vessel damage may lead to autoregulatory failure, making them more vulnerable to cerebral hypoperfusion and less tolerant of systemic BP reductions.[12,18–20] Moreover, global cerebral autoregulatory impairment observed in the early days after EVT may further exacerbate this risk.[24] Actually, increased WMH burden has been associated with worse outcomes in patients undergoing EVT, without an increase in the risk of hemorrhagic transformation.[25–27] Supplemental findings of our study are consistent with this pathophysiological framework: intensive BP lowering was associated with the greatest functional harm in those with the highest total WMH scores (5–6), with a nearly 96% reduction in the odds of achieving functional

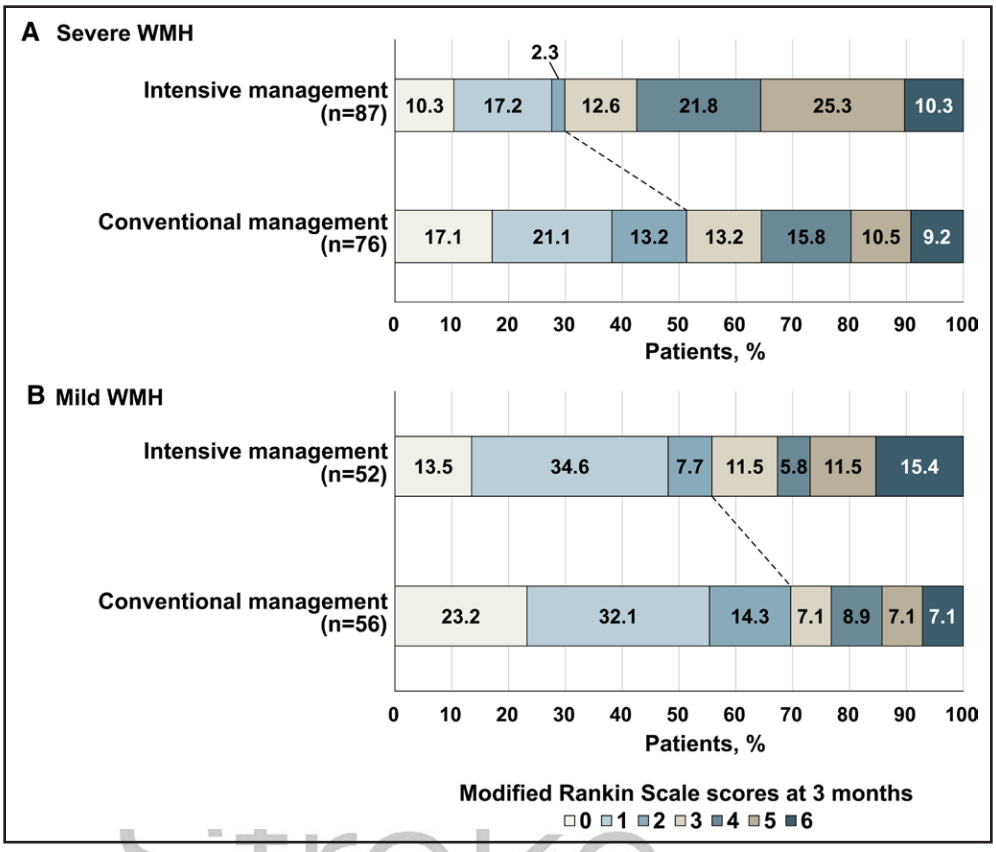

Figure 2. Distribution of modified Rankin Scale (mRS) score at 3 months according to severe white matter hyperintensity. WMH indicates white matter hyperintensity.

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independence. A similar, though nonsignificant, trend in the intermediate WMH group (3–4) suggests a possible dose-response relationship. These results indirectly support the theoretical mechanism that intensive BP management may exacerbate ischemic injury after EVT, with a differential impact depending on WMH burden and particularly worse outcomes in patients with severe WMH. Although infarct volume was significantly larger in the intensive management group compared with the conventional group among patients with mild WMH, this difference did not translate into worse neurological outcomes. As infarct volume was assessed as a secondary outcome and small sample size with large CI, the results should be interpreted with caution, and further studies are warranted.

Our supplemental analyses further support this interpretation. Intensive BP management was significantly associated with worse outcomes in patients with a more severe burden of periventricular WMH (score 2–3), but not in those with milder periventricular WMH (score 0–1). In contrast, intensive BP management was not associated with worse outcomes in patients with deep WMH scores of 2–3. The periventricular region, a watershed zone supplied by terminal arterioles of the medullary circulation, is particularly susceptible to cerebral hypoperfusion and ischemic injury, whereas deep WMH is associated with small vessel disease and has been linked to vascular dementia.[28] A decline in total cerebral blood flow has been associated with an increase in periventricular WMH burden (OR, 1.32 [95% CI, 1.06–1.66]; P=0.015), but not with deep WMH burden (P>0.05).[29] Further region-specific analyses have shown that periventricular WMH, but not deep WMH, was associated with cerebral hypoperfusion.[20,30] Moreover, in terms of clinical outcomes, previous studies have demonstrated that both total WMH burden and periventricular WMH burden were associated with poor functional outcomes after EVT, whereas deep WMH was not.[26,31] Interestingly, an unexpected pattern was observed in the deep WMH subgroups: patients with deep WMH scores of 0 to 1 experienced worse outcomes under intensive BP management, opposite to the trend observed with total WMH burden. Although the reason for this unexpected association is unclear, given that this finding runs counter to the overall trend, we believe it is most appropriate to interpret it as a chance finding or an exploratory observation. One possible explanation is that WMH burden may be linked to specific topographical patterns of stroke that disproportionately affect functional outcomes,[32] even when the preexisting WMH burden in deep regions is minimal. These results should be interpreted with caution and warrant further investigation.

Table 2. Clinical Outcomes by Management Group According to WMH Severity

	Severe WMH				Mild WMH				
	Intensive management (n=87)	Conventional management (n=76)	Adjusted effect size (95% CI)*	*P* value	Intensive management (n=52)	Conventional management (n=56)	Adjusted effect size (95% CI)*	*P* value	*P* for interaction
Primary efficacy outcome									
Functional independence at 3 mo (mRS score 0–2), n/total (%)	26/87 (29.9)	39/76 (51.3)	0.36 0.16–0.79)	0.011	29/52 (55.8)	39/56 (69.6)	0.77 (0.31–1.95)	0.584	0.355
Primary safety outcomes									
Symptomatic ICH, n/total (%)	7/87 (8.0)	2/76 (2.6)	2.90 (0.55–15.18)	0.207	4/52 (7.7)	4/56 (7.1)	0.51 (0.08–3.30)	0.478	0.277
Death related to index stroke within 3 mo, n/total (%)	5/87 (5.7)	4/76 (5.3)	1.47 (0.28–7.81)	0.650	4/52 (7.7)	1/56 (1.8)	1.21 (0.17–879.13)	0.255	0.419
Secondary outcomes									
mRS score reduction (shift analysis)			0.50 (0.28–0.88)	0.015			0.87 (0.42–1.77)	0.695	0.456
Excellent recovery in NIHSS score at 24 h, n/total (%)	9/86 (10.5)	15/76 (19.7)	0.55 (0.21–1.38)	0.202	16/52 (30.8)	22/56 (39.3)	0.88 (0.37–2.08)	0.762	0.501
Functional independence at 1 mo (mRS score 0–2), n/total (%)	26/85 (30.6)	33/74 (44.6)	0.63 (0.30–1.34)	0.229	25/50 (51.0)	37/56 (66.1)	0.70 (0.27–1.78)	0.448	0.976
EQ-5 d-3 L score, median (IQR)	0.4 (0.0–0.9)	0.7 (0.1–0.9)	−0.12 (−0.23 to −0.01)	0.037	0.8 (0.3–1.0)†	0.9 (0.7–1.0)‡	−0.05 (−0.19 to −0.08)	0.444	0.380
Infarct volume, median (IQR), mL	15.0 (5.3–56.9)	15.6 (5.2–35.5)	−1.11 (−26.87–24.65)	0.932	19.8 (9.9–85.9)	12.9 (5.4–36.6)	29.69 (0.07–59.32)	0.049	0.126

EQ-5 d-3 L indicates EuroQoL 5-Dimension, 3-Level Questionnaire; EVT, endovascular thrombectomy; ICH, intracerebral hemorrhage; IQR, interquartile range; mRS, modified Rankin Scale; mTICI, modified Thrombolysis in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; and WMH, white matter hyperintensity.

*Treatment effects are reported as proportional ORs (95% CIs) for the mRS shift analysis (ordinal logistic regression); adjusted mean differences (95% CIs) for the EQ-5 d-3 L scores and infarct volume (linear regression); and ORs (95% CIs) for other binary outcomes (binary logistic regression). All models were adjusted for age, sex, time from onset to reperfusion, IV thrombolysis, mTICI 3 recanalization, and NIHSS score immediately before EVT, in line with the OPTIMAL-BP trial design with minor modifications.[7]

†n=49.

‡n=55.

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This study has several limitations. First, it is a secondary analysis not prespecified in the original clinical trial design; therefore, causal inferences should be made with caution. Second, WMH burden was assessed visually using FLAIR imaging and categorized semi-quantitatively; more precise volumetric analyses may improve sensitivity. Although a substantial proportion of patients underwent FLAIR only after EVT, we mitigated potential bias through an independent core laboratory where blinded raters strictly cross-referenced FLAIR with diffusion-weighted imaging to differentiate chronic WMH from acute ischemic lesions, ensuring high diagnostic accuracy and excellent inter-rater reliability. Additionally, onset-to-FLAIR timing was comparable between BP management groups, minimizing temporal confounding. Third, direct measurements of cerebral autoregulation or hypoperfusion were not available, limiting mechanistic interpretation. Fourth, the relatively small sample size in each WMH subgroup may have limited the statistical power to detect modest associations or interactions. Finally, this study shares the inherent limitations of the original OPTIMAL-BP trial.[7] Nearly half of the patients treated with EVT were excluded because their postprocedural systolic BP was <140 mm Hg; thus, our findings cannot be generalized to all EVT patients. The early termination of the trial may have reduced statistical power and increased the likelihood of random or large treatment effects. In addition, the study population was limited to Korean patients, which may limit the generalizability of our findings.

Despite these limitations, this is the first study to evaluate the role of WMH guiding BP management after successful EVT. Our findings raise important considerations for post-EVT BP management. WMH burden, which can be readily assessed through routine MRI, particularly using the easily applicable Fazekas scale,[15] may serve as a practical and scalable marker, potentially acting as an effect modifier to individualize BP targets. Previous study has attempted to identify imaging markers such as Alberta Stroke Program Early CT Score, collateral flow status, and expanded TICI scores for BP management, but none have focused on WMH.[33] Future studies should incorporate physiological monitoring, perfusion imaging, and stratified randomization based on small vessel

disease burden, and should validate these findings in larger, adequately powered prospective cohorts.

CONCLUSIONS

In conclusion, our study suggests that intensive BP management after EVT may impair functional recovery, particularly in patients with severe WMH. Given its potential to increase susceptibility to BP reduction, WMH burden may serve as a marker for identifying patients who could benefit from more permissive BP targets in the early post-EVT period. These findings support a shift toward personalized BP management strategies informed by markers of small vessel disease and cerebrovascular reserve.

ARTICLE INFORMATION

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Author Contributions

Drs Baik, Jung, Jeon, Lee, and Nam performed data analysis. Drs Baik, Jung, Lee, Heo, Kim, Joo, Kim, Kim, Shin, Cho, Ahn, J. Yun, Park, Sohn, Hong, Song, Chang, Kim, Chang, Seo, Lee, Baek, Cho, Shin, Kim, J. Yoo, Seo, Jung, Hwang, Kim, Kim, Lee, Choi, Kim, Lee, Park, Kwon, Bang, Heo, and Nam obtained local data. Drs Baik, Jung, and Nam wrote the first draft of the article or prepared the figures. All authors reviewed and edited the article and approved the final version of the article.

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Supplemental Material

STROBE Checklist
Tables S1–S2
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APPENDIX

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